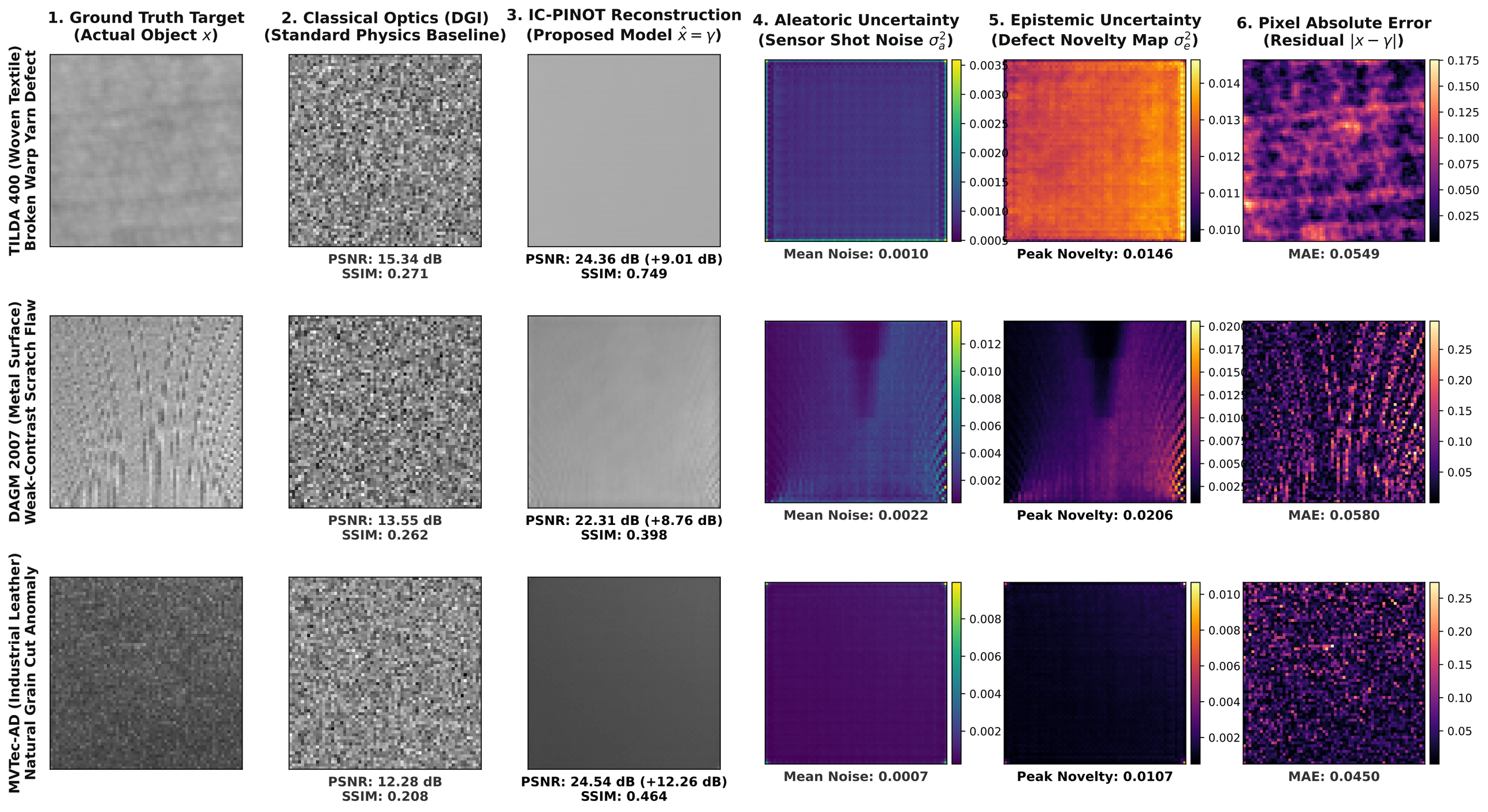

1. Ground Truth Target (Actual Object x)
2. Classical Optics (DGI) (Standard Physics Baseline)
3. IC-PINOT Reconstruction (Proposed Model $\hat{x} = \gamma$)
4. Aleatoric Uncertainty (Sensor Shot Noise σ_a^2)
5. Epistemic Uncertainty (Defect Novelty Map σ_e^2)
6. Pixel Absolute Error (Residual $|x - \gamma|$)
TILDA 400 (Woven Textile) Broken Warp Yarn Defect
PSNR: 15.34 dB
SSIM: 0.271
PSNR: 24.36 dB (+9.01 dB)
SSIM: 0.749
Mean Noise: 0.0010
Peak Novelty: 0.0146
MAE: 0.0549
DAGM 2007 (Metal Surface) Weak-Contrast Scratch Flaw
PSNR: 13.55 dB
SSIM: 0.262
PSNR: 22.31 dB (+8.76 dB)
SSIM: 0.398
Mean Noise: 0.0022
Peak Novelty: 0.0206
MAE: 0.0580
MVTec-AD (Industrial Leather) Natural Grain Cut Anomaly
PSNR: 12.28 dB
SSIM: 0.208
PSNR: 24.54 dB (+12.26 dB)
SSIM: 0.464
Mean Noise: 0.0007
Peak Novelty: 0.0107
MAE: 0.0450